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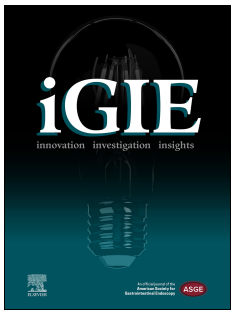


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# Mixed-Reality ESD Simulator: First Western Experience

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# Title:

Face validity of a new mixed-reality simulator for training endoscopic submucosal dissection: a Western pilot study.

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**Keywords:**

Endoscopic Submucosal Dissection (ESD), Virtual Reality (VR) Simulation, EndoCubot, Gastrointestinal Endoscopy, Medical Simulation Technology

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**Abstract**

**Background and Aims:** Endoscopic Submucosal Dissection (ESD) is an effective but technically demanding procedure with a steep learning curve. Traditional training methods, including master courses, apprenticeship and animal models, have ethical, financial, and logistical constraints. This pilot study evaluates the face validity of a novel mixed-reality simulator designed to enhance and maintain endoscopic dexterity in ESD training.

**Methods:** In this exploratory, prospective observational cohort study, eight ESD trainees with prior advanced endoscopy experience, but limited ESD exposure, underwent a two-day structured training using the EndoCubot® (EndoRobotics, Seoul, South-Korea) simulator with artificial laminated tissue (Versatile Training Tissue, VTT®,KOTOBUKI Medical, Yashio, Japan). Performance was assessed pre- and post-training through a 5-minute dexterity test simulating circular incision with 40 achievable points, and a questionnaire evaluating simulator realism, satisfaction, and training effectiveness. Paired samples t-tests ($\alpha = 0.05$) and descriptive statistics were used for analysis. Results were reported as mean with standard deviation (SD).

**Results:** On a 40-point scale mean post-course scores (34.7 (4.86)) were significantly higher than pre-course scores (25.50 (0.76)), $p < 0.001$), with a mean improvement of 9.25 points (95% Confidence Interval (CI): 4.993–13.507, Hedges' g correction = 1.613). Survey results indicated high satisfaction (mean rating: 9.125/10) and strong agreement on the simulator's realism and effectiveness.

**Conclusions:** The simulator demonstrates strong face validity and supports a realistic standardized skill acquisition in ESD. The simulator could provide a realistic and ethical alternative to current animal models and requires further study.

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## Introduction

Endoscopic submucosal dissection (ESD) is an advanced minimally invasive procedure to remove early-stage tumors from the superficial layers of the gastrointestinal tract. The technique enables an en-bloc resection with accurate pathology staging. It is a tedious technically demanding procedure and requires extensive knowledge, specialized equipment and a high level of new endoscopic skill. Even for experienced endoscopists, ESD is characterized by a steep learning curve and possible loss of dexterity due to lack of continuous practice. Western ESD training methods often rely on hands-on practice using explant models and live porcine models.[1] These models are limited by availability, ethical concerns, costs, anatomical discrepancies, and logistical hurdles related to specimen procurement, maintenance, and endoscope hygiene.[2–4]

Over the last decades multiple endoscopy training simulators have been constructed and evaluated.[2,3] These simulators offer a risk-free, reproducible environment for repetitive practice, allowing trainees to hone their skills before performing procedures on patients. Several studies have demonstrated that computer-assisted models (e.g., virtual-reality platforms) can effectively supplement conventional training by accelerating skill acquisition and enhancing procedural competency.[3–5] Moreover, development of mixed-reality models (i.e simulators that blend real-world and virtual elements) using artificial tissue have improved the realism and educational value of simulation-based training.[6]

The EndoCubot simulator (EndoRobotics, Seoul, South-Korea) first introduced in 2024 represents a significant advancement by bringing everyday real instruments into a virtual-reality (VR) platform with a dynamic positioning system that mimics gastrointestinal conditions. Unlike conventional models, EndoCubot enables precise manipulation of standard endoscopic instruments while simulating different anatomical lesion positions, physiological movements (e.g., respiration and peristalsis), and distinct lesion architecture. By eliminating the need for live animal models, it addresses ethical concerns, reduces training costs, and might assist in ensuring standardized skill acquisition and dexterity improvement.

This is the first pilot study aimed to assess the content- and face validity of the EndoCubot simulator as a novel training tool for ESD evaluating its impact on endoscopic dexterity.

## Materials and Methods

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*EndoCubot Simulator*

The EndoCubot (EndoRobotics, Seoul, South-Korea) is an innovative mixed-reality positioning simulator specifically designed for ESD training (Figure 1). Departing from traditional reliance on animal or ex-vivo specimens, it offers a controlled, reproducible training environment that closely mimics various clinical endoscopic scenarios. The simulator utilizes artificial laminated tissue (Versatile Training Tissue, VTT®; Kotobuki medical, Japan) to replicate the architecture of the gastrointestinal wall, enabling high-fidelity practice. Its features include adjustable wall positioning, simulated tissue movements (including respiratory and peristaltic motions), and dynamic feedback to improve endoscopic dexterity. The EndoCubot allows simulation of a variety of gastrointestinal anatomies ranging from the esophagus, stomach, colorectum, but also more complex regions such as the cardia in retroflexion and gastric angle. **(Figure 1C)** Additionally, EndoCubot delivers objective performance metrics, facilitating data-driven assessments of trainee progression and ensuring standardized training.

*Study Design*

A prospective observational cohort pilot study was conducted to evaluate the face validity of the EndoCubot simulator for ESD training. Improvements in endoscopic dexterity and trainee perceptions were assessed using standardized performance metrics and questionnaire-based evaluations.

*Participants*

Eight ESD trainees were recruited from interventional endoscopy expert-centers across Europe. Inclusion criteria required prior experience in basic upper and lower endoscopic procedures (biopsies, hemostasis, polypectomies, mucosal resections) and limited exposure to clinical ESD (<20 procedures).

As this pilot feasibility study did not involve research on patients, no local Institutional Review Board approval was needed. Therefore, only oral informed consent was obtained from all participants/health care professionals. Participants involved were free to answer the questionnaires following the conclusion of this training.

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*Dexterity Training*

The simulator-based training was delivered under guidance of three experts over two days (minimum 2.5 hrs training time per trainee/day) in February 2025 using the EndoCubot simulator with artificial laminated tissue (VTT). Procedures were performed with a standard 4mm injection needle, a standard DualKnife J® (Olympus, Tokyo, Japan) and an Olympus ESG-300® electrosurgical generator with standard ESD settings. No equipment failures were observed. The curriculum employed a structured, stepwise approach that progressively increased task complexity, i.e. lesion and position complexity as well as tissue movements. Trainees engaged in standardized ESD exercises -including lesion marking, submucosal injection, circumferential cutting, and en-bloc dissection- to build conventional ESD specific endoscopic dexterity. The anatomic positions used for training were greater curvature aspect of the antrum, anterior wall of the antrum, incisura angularis, gastric cardia in retroflexion, as well as proximal rectum and colon.

*Assessments and Evaluation*

*Dexterity Test:* An identical 5-minute test using the antrum posterior wall position of the simulator was administered at both the start and conclusion of the training. The test developed for this study involves the marking and cutting of two circular 3 centimeter (1.18 inch) and 4.5 centimeter (1.77 inch) lesions on the VTT in clockwise or counterclockwise fashion (Figure 2). Performance within the allotted time was quantified based on the accuracy of lesion marking -each successfully placed mark is awarded with 1 point-, and completeness of circumferential cutting – where each complete connecting cut between two marks is awarded with 1 point. Thereby a maximum performance score of 40 points can be obtained.

*Questionnaire-Based Evaluation:* Post-training, validity-based evidence was obtained through a subjective survey-based analysis. The survey designed for this study included 14 statements evaluating training effectiveness, simulator performance and comparison with porcine training models. A 5-point Likert scale was used to assess participants' perceptions of the simulator and the training program.

*Statistical Analysis*: Data were analyzed using IBM SPSS Statistics version 30.0 (IBM, Armonk, NY, USA). A paired sample t-test was conducted to assess differences between pre-course and

post-course performance scores. Effect sizes were calculated using Hedges' g correction to quantify the magnitude of observed differences. Boxplot analysis was performed to visualize performance score distributions before and after training.

Descriptive statistics, including medians, ranges, interquartile ranges, and means were calculated for survey responses. All statistical analyses were performed two-sided using a significance threshold of $\alpha=0.05$.

Results were reported as mean (standard deviation (SD), median (interquartile range [IQR], and 95% Confidence Interval (CI)).

Ethical Considerations: All participants were assured that study data would remain anonymous, confidential and be managed in strict accordance with the protocol. No data were shared with third parties or contracting authorities.

**Results**

*Participant Demographics:* The study sample consisted of 8 participants (2 females and 6 males) from 8 European expert centers, all of whom were ESD novices but had finished endoscopic training in their country of origin. On average, participants had attended 1.6 prior ESD training courses (SD 1; range: 0–3) and had performed an average of 7.8 ESD procedures (SD 7; range: 0–20).

*Performance score outcomes:* Mean post-course scores 34.7 (4.86) out of a maximum score of 40 points were significantly higher than the pre-course scores (25.50 (0.76)); $p<0.001$). Analysis yielded a mean difference of 9.25 points, with the 95% confidence interval ranged from 4.99 to 13.51 points (Figure 3). Effect size calculations with Hedges' g correction at 1.613 further supported this meaningful improvement in performance.

*Survey Findings*:

Subjective evaluations of the EndoCubot simulator and training course were collected using a 5-point Likert scale. The EndoCubot simulator received high marks. Analysis of the questionnaire responses showed that most of the 14 statements regarding training and

simulator performance were rated at the upper end of the scale (medians of 4–5) (Table 1). Specific aspects of the training program -including the quality of the learning environment, appropriateness of the time allocated on the simulator, and the usefulness of instructor feedback- received consistently high ratings. Similarly, items related to the technical functionality of the simulator (e.g., tissue simulation quality, realistic training scenario simulation, and wall movement representation) were also rated highly. Furthermore, participants expressed a preference for this simulator over traditional animal-based models, with positive responses regarding its utility.

**Discussion**

ESD is a technically demanding procedure that requires a high level of endoscopic skill and hands-on experience. Generally, a minimum of 20 hours hands-on practice in a learning environment is advised. Studies suggest that subsequently more than 30 supervised clinical cases are necessary to achieve proficiency and reduce adverse events.[1] However, access to expert mentorship and structured master training programs remain limited globally, posing significant barriers to standardized ESD education.[1,7] Furthermore, dexterity deficit and dexterity decay undermine the psychomotor component of ESD also in more experienced hands. In high-volume centers abundant opportunities exist for effective practice and dexterity improvement, however the average clinical environment is frequently not conductive to effective training practice.

To address this, various training models have been developed outside of frontline medical practice. The most widely used are porcine models. These models have demonstrated benefit in tutored learning environments by accelerating the learning curve. Despite their relative realism, porcine models present considerable limitations such as tactile differences (e.g., mucosal thickness with high tissue resistance), ethical concerns, high costs (both financially and ecologically), logistical demands (including biosafety, anesthesia and animal handling), and access to veterinary and technical support as well as dedicated endoscopes for animal use.[8]

VR simulators, such as Endosim® (Endosim USA, Bolton, MA, USA) have demonstrated strong face validity and efficacy for basic endoscopy training but have limitations in advanced endoscopy training.[3–5,9–12] They lack tactile realism, including elasticity, tissue resistance,

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and haptic feedback, and often simulate instrument handling in oversimplified environment limiting their suitability for ESD training and skill maintenance. Conversely, a recent study on a PVA-H (Polyvinyl Alcohol Hydrogel) based ESD simulator found that repeated training sessions led to improved complete resection rates and reduced perforation rates. This supports the role of high-fidelity simulation in progressive skill development before transitioning to live procedures.[6]

The EndoCubot used in this small exploratory pilot experiment potentially addresses this gap by combining actual endoscopic instruments with a dynamic positioning system that mimics human-physiology and anatomy, including folded lesions, lesion movement, tissue resistance and variable lesion locations. Unlike animal or merely virtual models, it can provide a controlled, reproducible, and ethical environment for practicing, repeating, and assessing ESD-related skills.

In our pilot study, the EndoCubot demonstrated strong face validity, with a 36.3% mean improvement in performance scores after a short, structured training program. This number may underestimate the benefit, as 25% of participants completed the post-test well within the allocated time, indicating enhanced efficiency. However, in this exploratory study we could not assess how much of this improvement is transferable to real-life ESD procedures in patients, an important limitation shared with most simulation-based studies.

Beyond objective performance, the trainees subjectively rated the simulator highly across multiple domains, including realism, usability, and perceived effectiveness. Given their background in upper and lower gastrointestinal therapeutic endoscopy, and prior exposure to training courses, the trainees seem well fitted to judge the simulator's realism and relevance. The partial task training concept and the learning environment both appear to represent the intended purpose, i.e., the opportunity for the rehearsal of psychomotor skills under various circumstances in a risk-free controlled environment. Secondly, the median global classifications of the items related to the realism of the simulator model suggests that the model mimics actual circumstances and that procedures accurately resemble elements of clinical ESD. In the survey, the wall structure of the plant-based tissue scores are respectable, but attained the lowest value compared to other aspects of the simulator. With respect to the comparison of the model with the current animal models, a preference for the mixed-reality

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computer simulator is observed. This could be related to the advantage of controlling the difficulty level and unconstrained repetition that makes this model approach the ideal teaching tool. Yet, ethical and emotional considerations are likely to be also involved.

While the findings highlight the benefits of simulation-based ESD training, this study has several limitations such as its small sample size, pilot study design and short observation period that may not be able to depict long-term skill retention or sustained competency development with more complex interventions simulated. Also, the trainees might have rated the simulator favorably due to perceived expectations rather than actual training outcomes as an expression of survey bias. In addition, a direct comparison with a live model training is lacking.

Future Western studies should explore cost-effectiveness analyses in comparison to animal models to determine its feasibility as a large-scale training solution.

This pilot study adds to the growing evidence supporting simulation-based training for ESD. These preliminary results indicate that the EndoCubot simulator could provide a realistic, standardized, ethical and ecological alternative to animal models, demonstrating measurable improvements in trainee performance and high student satisfaction. These findings could suggest that structured simulator training can play a critical role in advanced endoscopy skill acquisition and skill maintenance. Future research should establish validated competency benchmarks for ESD training and study whether simulation-based training aids trainees in achieving these benchmarks.

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**Figure 1 A-C: ESD (Endoscopic submucosal dissection) training simulator from**

**EndoRobotics Co.** (Seoul, South-Korea)

A: Overall training setup with the Endocubot (marked with arrow) and Endoscopy tower

B: Internal machine mechanism with scope tray and artificial tissue holder

(green arrow). Arrows show the possible movements of the scope tray and specimens

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C: Possible anatomical settings of artificial tissue with various holder trays that get inserted into the Enocubot for simulation of different anatomical scenarios for resection

Figure 2. Assessment involving the marking and cutting of two circular lesions 3 cm (1.18 inches; yellow) and 4.5 cm (1.77 inches; blue) in diameter on the plant-based tissue.

Figure 3. Boxplot of pre- and post-course performance score results expressed as mean Standard deviation (SD) 25.50(0.76) vs. 34.75 (4.86) p<0.001).

Figure 4. Example of the paired dexterity assessment in the antrum posterior wall position at the start with only about half of the blue dots connected (left) and conclusion of the training with all yellow and blue dots connected correctly (right).

| | **Median** | **Min /Max** | **IQR** | **Mean** |
|---|---|---|---|---|
| **Training programme** | | | | |
| Learning environment was satisfactory | 5 | 4/5 | 0 | 4.86 |
| Time allocated on the simulator was satisfactory | 4 | 4/5 | 1 | 4.43 |
| Training feed-back by instructors | 5 | 4/5 | 1 | 4.71 |
| Presence of other trainees was not hindering | 5 | 4/5 | 1 | 4.57 |
| Observing other trainees perform was helpful | 5 | 4/5 | 0 | 4.86 |
| **EndcoCubot simulator** | | | | |
| Simulator functioned technically well | 5 | 4/5 | 1 | 4.71 |
| Simulator in this training is useful | 5 | 4/5 | 1 | 4.71 |
| Quality of tissue simulation | 4 | 3/5 | 1 | 4.29 |
| Simulates real-life training scenarios | 5 | 3/5 | 1 | 4.43 |
| Simulated wall movements are helpful | 4 | 4/5 | 1 | 4.43 |
| I would like more ESD training with simulators | 5 | 4/5 | 1 | 4.57 |
| **Simulator in comparison** | | | | |
| Simulator useful addition to training on patients | 5 | 4/5 | 1 | 4.71 |
| Simulator is better than the pig stomach model | 4 | 3/5 | 2 | 4.00 |
| Simulator is preferred over live pigs | 4 | 3/5 | 2 | 4.14 |

Table 1. 14 post-training evaluation statements grouped by topic and rated on a Likert scale (1=strongly disagree, 2=disagree, 3=neutral, 4=agree, 5=strongly agree)
Minimum score given (Min); Maximum score given (Max), Interquartile Range (IQR), Endoscopic submucosal dissection = ESD

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| Procedure | Tissue Holder | Tissue Attached | Endoscopic View |
|---|---|---|---|
| Stomach General | | | |
| Stomach Incisura Angularis | | | |
| Colorectal | | | |
| Esophagus | | | |

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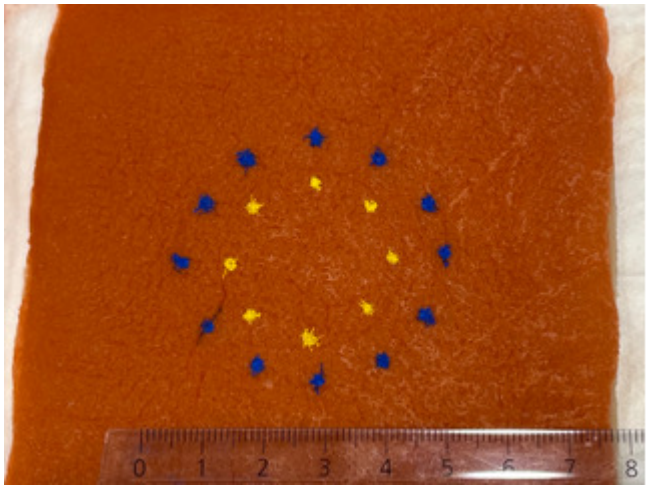


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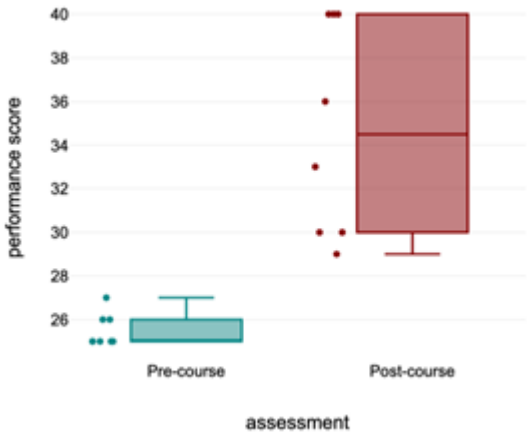


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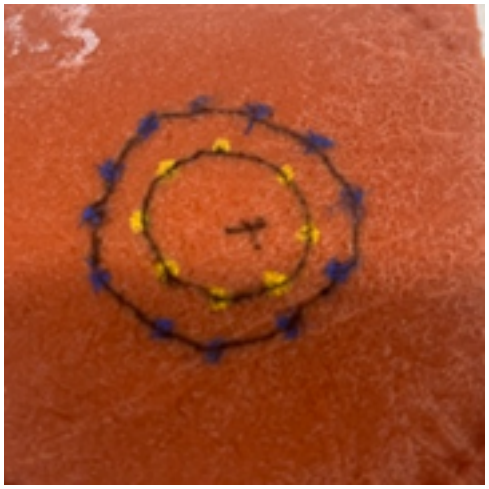

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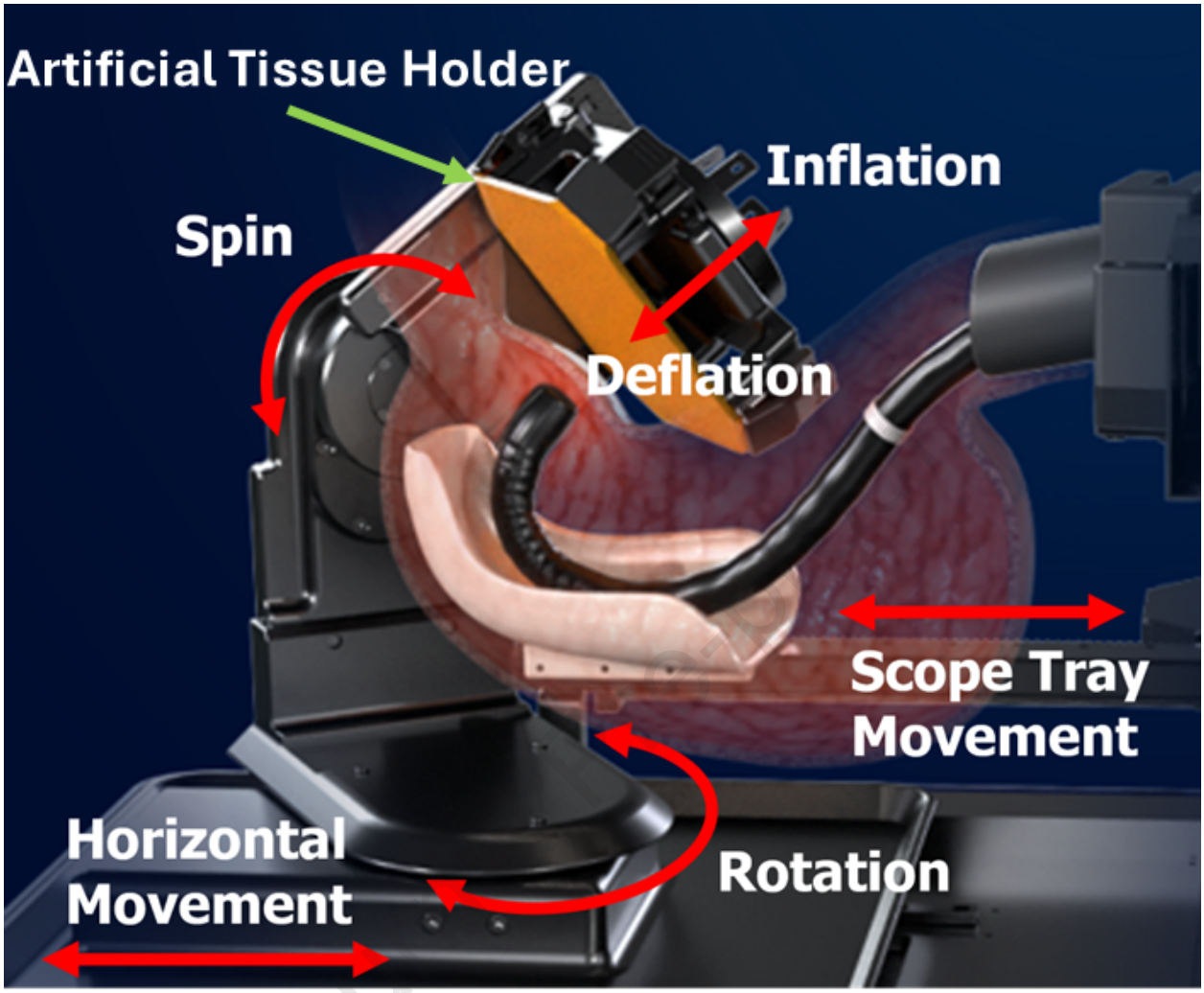
Artificial Tissue Holder
Inflation
Spin
Deflation
Scope Tray
Movement
Horizontal
Movement
Rotation

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Abbreviations and Accronyms

CI Confidence Interval

ESD Endoscopic Submucosal Dissection

GI Gastrointestinal

IRB Institutional Review Board

VR Virtual Reality

VTT artificial laminated tissue

SD Standard Deviation

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